Homework 5

PHY180H1F Fall 2026

Online Submission of Questions 1-15: Due Monday Oct.19 by 11:59pm

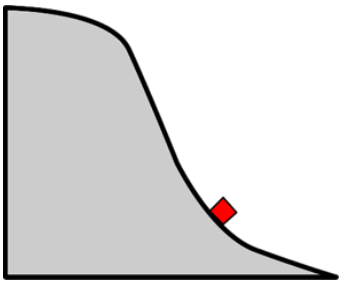
Instructions

The answers to Questions 1 through 15 in this Homework must be submitted by the deadline on Quercus at https://q.utoronto.ca/courses/469387 by entering the numerical answer, or, sometimes, multiple choice. I encourage you to print this PDF and solve the problems with pen and paper, making sure you *understand your own reasoning!* When you are ready, go to Quercus to enter what you get and check your work. You get up to 4 attempts at each question, with full credit if any of the attempts is correct.

Many questions are end-of-chapter "Questions" or "Problems" from "University Physics Volume 1", by Moebs, Ling and Loyola, ©2026 by Rice University, where the first number is the chapter, and the decimal is either the Conceptual Question number with a Q in front, or the Problem & Exercises number with a P in front. For example, if it says "UPV1 OpenStax Q3.22", it means Chapter 3, Conceptual Question 22. If it says "UPV1 OpenStax P10.95", it means Chapter 10, Problem number 95.

For all questions, please assume that air resistance is negligible, unless otherwise stated.

Question 1. A cart slides without friction down a track as shown. As the cart slides beyond the point shown, what happens to its acceleration and its speed, both in the direction of its motion?



A. Both decrease.
B. The speed decreases, but the acceleration increases.
C. Both remain constant.
D. The speed increases, but the acceleration decreases.
E. Both increase.

Chapter 8: Potential Energy and Conservation of Energy

https://openstax.org/books/university-physics-volume-1/pages/8-problems :

Question 2. UPV1 OpenStax P8.28. Assume all motion is along the *x*-axis.

28. A particle of mass $2.0\ \text{kg}$ moves under the influence of the force $F(x) = (3/\sqrt{x})\ \text{N}$. If its speed at $x = 2.0\ \text{m}$ is $v = 6.0\ \text{m/s}$, what is its speed at $x = 7.0\ \text{m}$?

Question 3. UPV1 OpenStax P8.62

62. A 5.00×10^5 -kg subway train is brought to a stop from a speed of 0.500 m/s in 0.400 m by a large spring bumper at the end of its track. What is the spring constant *k* of the spring?

Question 4. UPV1 OpenStax P8.88

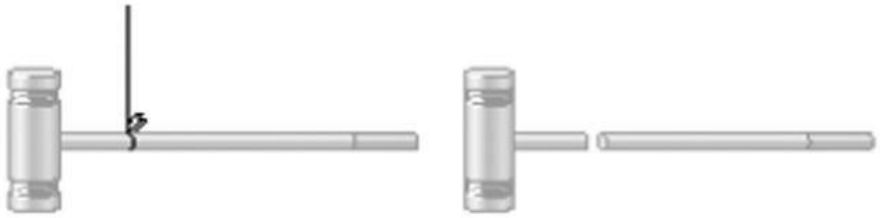
88. How much energy is lost to a dissipative drag force if a 60-kg person falls at a constant speed for 15 meters?

Question 5. An 80.0 kg bungee jumper drops from rest off a bridge that is 200.0 m above a river. As he falls, air resistance is negligible, and the unstretched bungee cord attached to him uncoils. When the jumper is 50.0 m below the bridge, the bungee cord is completely uncoiled and forms a straight line. Below that point, the bungee cord acts as a spring, obeying Hooke's law as it is stretched beyond its equilibrium length of 50.0 m, with a spring constant of k = 15 N/m. How far above the river will the jumper be when he first momentarily stops?

Question 6 Two sailboats have identical size and shape, but sailboat A has a significantly larger mass than sailboat B. They are both subject to the same wind and the same constant net force as they start from rest and move the same distance of 100 m. After traveling 100 m,

A. they both have the same momentum and kinetic energy.
B. they both have the same kinetic energy but sailboat A has larger momentum than sailboat B.
C. they both have the same kinetic energy but sailboat B has larger momentum than sailboat A.
D. they both have the same momentum but sailboat A has larger kinetic energy than sailboat B.
E. they both have the same momentum but sailboat B has larger kinetic energy than sailboat A.

Question 7. A croquet mallet balances when suspended from its center of mass, as shown in the left part of the figure. If you cut the mallet into two pieces at its center of mass, as shown in the right part of the figure, how do the masses of the two pieces compare?



A. The masses are equal.
B. The piece with the head of the mallet has the smaller mass.
C. The piece with the head of the mallet has the greater mass.
D. It is impossible to tell.

Chapter 9: Linear Momentum and Collisions

https://openstax.org/books/university-physics-volume-1/pages/9-problems :

Question 8. UPV1 OpenStax P9.42

42. A 2000-kg railway freight car coasts at 4.4 m/s underneath a grain terminal, which dumps grain directly down into the freight car. If the speed of the loaded freight car must not go below 3.0 m/s, what is the maximum mass of grain that it can accept?

Question 9. UPV1 OpenStax P9.46. Determine the speed.

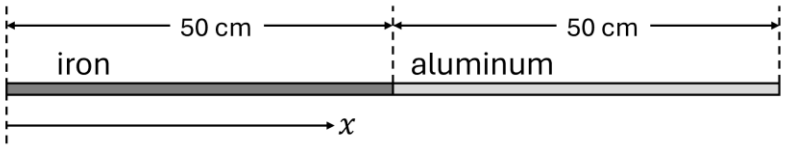
46. A 100-g firecracker is launched vertically into the air and explodes into two pieces at the peak of its trajectory. If a 72-g piece is projected horizontally to the left at 20 m/s, what is the speed and direction of the other piece?

Question 10. UPV1 OpenStax P9.62. Determine the speed.

62. Two cars of the same mass approach an extremely icy four-way perpendicular intersection. Car A travels northward at 30 m/s and car B is travelling eastward. They collide and stick together, traveling at 28° north of east. What was the initial velocity of car B?

Question 11. UPV1 OpenStax P9.68. Determine the x-position of the center of mass, with x defined by the diagram below.

68. Find the center of mass of a one-meter long rod, made of 50 cm of iron (density $8\ \frac{\text{g}}{\text{cm}^3}$) and 50 cm of aluminum (density $2.7\ \frac{\text{g}}{\text{cm}^3}$).



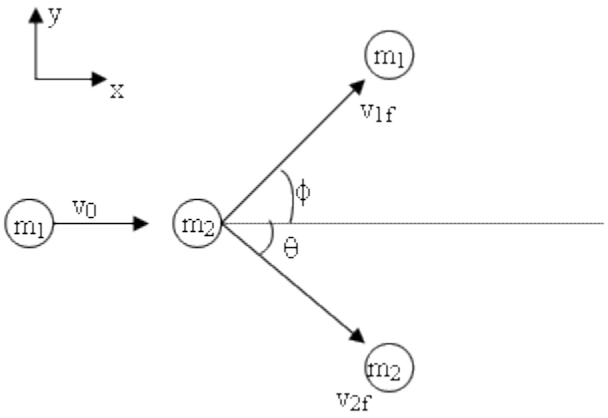
Question 12. UPV1 OpenStax P9.78

78. A rocket takes off from Earth and reaches a speed of 100 m/s in 10.0 s. If the exhaust speed is 1500 m/s and the mass of fuel burned is 100 kg, what was the initial mass of the rocket?

Question 13. A big wagon with a mass of 250 kg is initially rolling without friction at 3.00 m/s. It starts to rain, and the rain accumulates in the wagon at a constant rate of 10.0 kg/hr. The rain is falling straight down. After 2.00 hours of rain, what is the speed of the wagon?

Questions 14 and 15 are about the following situation:

A puck of mass m_1 = 0.2 kg is sliding on an ice table without friction. Its initial velocity is along the x-axis with v_0 = 10 m/s as shown in the figure. The puck collides elastically with an identical puck, m_2, which is initially at rest. The pucks exit the collision as shown in the figure.



It is observed that, after the collision, the velocity components of puck 2 are (v_{2fx}, v_{2fy}) = (8.6 m/s, −3.5 m/s).

Question 14. What is the *x*-component of the velocity of puck 1 after the collision?

Question 15. What is the *y*-component of the velocity of puck 1 after the collision?

Chapters 8 and 9 Extra-Practice Problems (Optional)

The Questions and Problems Below are for students who wish to do some *extra practice work*. They are **fully and completely optional,** and are NOT worth any marks.

Question 16. UPV1 OpenStax P8.29

Question 17. UPV1 OpenStax P8.31

Question 18. UPV1 OpenStax P8.63

Question 19. UPV1 OpenStax P8.72

Question 20. UPV1 OpenStax P8.87

Question 21. UPV1 OpenStax P9.29

Question 22. UPV1 OpenStax P9.47

Question 23. UPV1 OpenStax P9.54

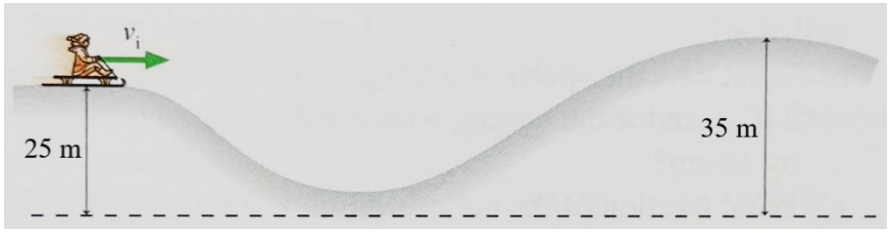
Question 24. UPV1 OpenStax P9.63

Question 25. UPV1 OpenStax P9.69

Question 26. UPV1 OpenStax P9.81

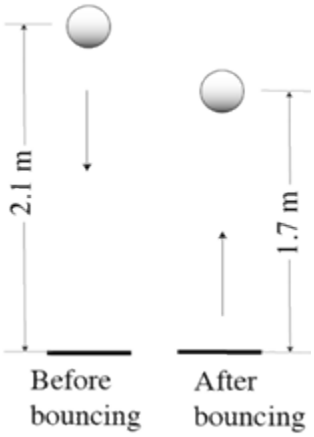
Question 27. UPV1 OpenStax P9.91

Question 28. The sledder shown in the figure below starts from the top of a frictionless hill and slides down into the valley. What initial speed v_i does the sledder need to just make it over the next hill?



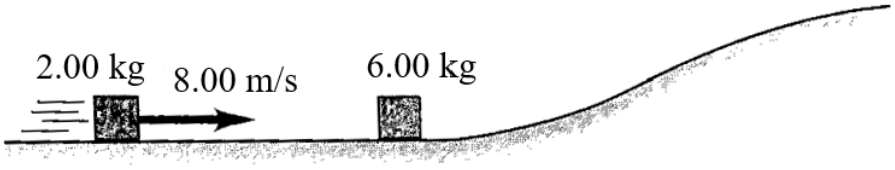
Question 29. A truck of mass m_T is moving with an initial speed v, and collides with a stationary car of mass m_C. The truck and car remain locked together and skid a distance d before stopping. The coefficient of kinetic friction between the wheels of the vehicles and the road is μ_k. What was the truck's initial speed?

Question 30. A ball falls vertically from the initial height of 2.1 m with no initial velocity to the floor and bounces vertically, reaching the highest point 1.7 m from the floor as illustrated. Take the origin of the potential energy of the ball to be that on the floor. The mass of the ball is 0.120 kg. What is the magnitude of the impulse on the ball from the floor when the ball bounces?



Questions 31 and 32 are about the following situation:

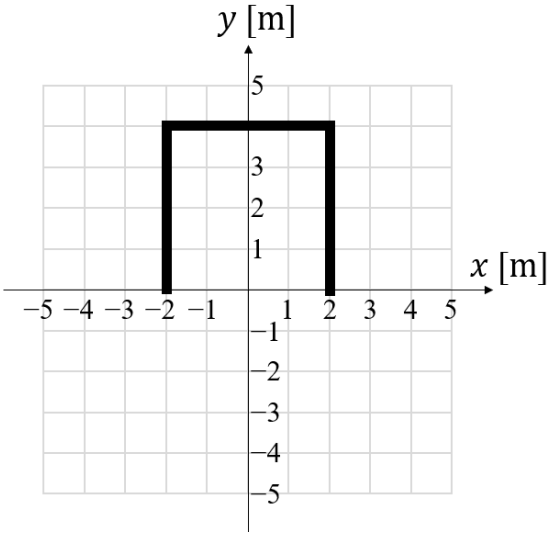
A 2.00 kg chunk of ice is sliding at 8.00 m/s on the floor of an ice-covered valley when it collides with a 6.00 kg chunk of ice that is initially at rest. The surface is frictionless everywhere.



Question 31. Assume the collision of the two chunks of ice is perfectly inelastic, meaning that they stick together. After the collision, how high above the valley floor will the combined chunks go?

Question 32. Assume, instead, that the collision of the two chunks of ice is head-on and elastic. This means that, during the collision, all motion is along a line, momentum is conserved, and also kinetic energy is conserved. After the collision, how high above the valley floor will the 6.00 kg block go?

Question 33. A 3.0 kg object is made of three long thin uniform metal bars, each of which has a mass of 1.0 kg and length 4.0 m. The bars are connected together as shown (thick black lines), and lie in the x-y plane. What are the (x,y) coordinates of the centre of mass of this object?



Appendix A: Answers To Practice Questions 16-33.

If you need help or explanation of how I got any of these answers, feel free to check out the YouTube video at https://youtu.be/rFDCKWE6snE

16. 14 m/s
17. −14 J
18. 0.34 m
19. [*proof question – see video*]
20. 90 m/s
21. 2100 N
22. First car: 6.00 m/s, second car: 5.60 m/s
23. 2.7 m/s, 8.1° away from original direction.
24. (2.8, 0.69) cm
25. $\frac{3}{4}L\left(\frac{\rho_0+\rho_1}{2\rho_0+\rho_1}\right)$
26. 4700 m/s
27. 0.021 kg m/s, upward
28. 14 m/s
29. $\frac{m_T+m_C}{m_T}\sqrt{2g\mu_k d}$
30. 1.5 kg m/s, upward
31. 0.204 m
32. 0.816 m
33. (0, 2.7) m